Peng-Ting (Dean) Kuo

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Education

University of Texas at Austin — *Austin, TX*
Ph.D. in Government — *2025 - 2030*
- Major field: Methodology; Minor: International Relations

National Taiwan University (NTU) — *Taipei, Taiwan*
MA in Political Science — *2021 - 2024*
- Overall GPA: **4.06/4.3**
- Thesis: The Impact of Electoral Reform on Legislators' Campaign Strategies: An Application of Large Language Models

National Taiwan University (NTU) — *Taipei, Taiwan*
BA in Political Science and Minor in Economics — *2017 - 2021*
- Last two years GPA: **4.11/4.3**

Peer-reviewed Publications

LLMs are Biased Evaluators But Not Biased for Fact-Centric Retrieval Augmented Generation. *Findings of the Association for Computational Linguistics: ACL 2025 (Findings of ACL 2025), pages 26669–26684. Vienna, Austria, Jul 27 - Aug 1, 2025. (with Yen-Shan Chen, Jing Jin, Chao-Wei Huang, Yun-Nung Chen)*

Working Papers

Democracy or Prosperity? A Cross-National Analysis of East Asian Countries. *R&R*. (*with Ronan Tse-min Fu, I-Lien Lee, Chia-Hsiang Han, Yu-Ning Chiu*)

Presented at 2023 Asian Barometer Survey Conference

Decoding Campaigning Strategies: A Machine Learning Approach to Understanding the Impact of Election Systems on Legislative Candidates. (*with Ronan Tse-Min Fu, Nich Lin*)

Presented at 2025 IPSA; 2024 MPSA, JAES; 2023 TPSA

More Trade, More Dispute? Measuring Alliance Effect on WTO Conflict. (*with Wayne Hsuan-Wei Lee*)

Presented at 2026 MPSA

The Growth of Executive Constraints: A Global, Historical Account. (*with Andrés Cruz, John Gerring*)

Professional Experience

Department of Government, University of Texas at Austin — *Austin, TX*
Research Assistant — *Aug 2025 - Present*
- Supervised by Prof. John Gerring
- Engineered end-to-end data pipeline from scratch for extracting and structuring institutional data from historical texts using multiple state-of-the-art commercial and open-sourced LLMs
- Optimized performance through experimentation with prompt engineering, IRT modeling, and pairwise comparison

Institute of Sociology, Academia Sinica — *Taipei, Taiwan*
Research Assistant — *Apr 2024 - Present*
- Supervised by Prof. Wayne Hsuan-Wei Lee
- Analyzed 600+ WTO Dispute Settlement Body cases and 8000+ documents using social network analysis uncovering key patterns and relationships between countries
- Designed data pipeline from scratch, including web scrapping, text pre-processing, Retrieval Augmentation Generation (RAG) system, and data analytics, improving data accessibility for scalable research

Department of Computer Science & Information Engineering, NTU — *Taipei, Taiwan*
Research Assistant — *Aug 2024 - Dec 2024*
- Supervised by Prof. Vivian Yun-Nung Chen
- Optimized research design and evaluated bias in LLMs (GPT, LLaMA) using the Retrieval-Augmented Generation (RAG) framework, incorporating social science experimental methodology
- Curated and analyzed 1000+ paired human-AI annotated datasets, developed visualization tools for bias assessment, contributing to a paper presented by the 2025 ACL Findings

Institute of Political Science, Academia Sinica *Taipei, Taiwan*

Research Assistant *Mar 2022 - Aug 2025*

- Supervised by Prof. Ronan Tse-Min Fu
- Engineered datasets and conducted statistical analysis on cross-national data from twelve East Asian countries using Python and R, contributed to a publication at the Asian Barometer Survey Conference 2023
- Developed and curated a 2-million-entry news database from China's People's Daily. Trained classification models using PyTorch and RoBERTa framework, achieving 96% accuracy in identifying international news
- Developed and maintained an events database on UK-Germany relations, consolidating a comprehensive resource and literature reviews for the book project and regularly reporting progress to the team

First Bank *Taipei, Taiwan*

Intern *Jul 2020 - Aug 2020*

- Conducted qualitative data collection and developed strategic recommendations to enhance digital bank user experience, earning recognition from managers
- Initiated and organized an experience-sharing session on project management for 50+ interns, strengthening intern-company relationships

Conference Presentation

MPSA (2024; 2026); IPSA (2025); TexMeth (2026); JAES (2024); International Conference on Parliamentary Studies (2024)

Invited Talk

Exploring an Interdisciplinary Academic Career in Data Mining and NLP *2024*

Department of Political Science, Soochow University

- Guest lectured on applications of NLP in political science research during graduate statistics courses
- Led hands-on Python programming workshops demonstrating NLP implementation

Honors and Awards

Graduate Fellowship in Law and Democracy, *UT Austin* ($2000) *2026*

Phi Tau Phi Scholastic Honor Society, *National Taiwan University* *2025*

TOEFL ITP Higher Education Research Scholarship, *TOEFL Chun-Hsin Limited* *2024*

Travel Grant for International Conference Participation, *National Science and Technology Council; NTU* *2024*

Diplomatic Scholarship, *Ministry of Foreign Affairs, Republic of China (Taiwan)* *2022*

Yu Ren-Huan Scholarship, *Department of Political Science, NTU* *2019*

Dean's List Award (Rank 2/46), *NTU* *2019*

Teaching Experience

University of Texas at Austin *Austin, TX*

Statistical Analysis in Political Science (GOV350), TA *Spring 2026*

World War I in Real Time (GOV355), TA *Fall 2025*

Academia Sinica *Taipei, Taiwan*

Institution of Social Science Methodology (ISSM), TA *Summer 2025*

National Taiwan University (NTU) *Taipei, Taiwan*

Appied Deep Learning (CSIE5431), TA *Fall 2024*

Computational Social Science (PS5697), TA *Spring 2024*

Basic Mathematics for Social Science (PS5710), TA *Fall 2023*

Social Network and Group Behavior (PS5705), TA *Spring 2023*

Seminar on Game Theory (PS5019), TA *Fall 2022*

International Law (PS3316, PS3317), TA *Fall 2021 - Spring 2024*

Other Training

Summer Institute in Computational Social Science *Rochester, NY*

University of Rochester *Summer 2026*

Institute for Political Methodology *Taipei, Taiwan*

Academia Sinica *Summer 2022*

Skills

Language Mandarin, English, Japanese

Programming Python, PyTorch, R, Netlogo

Software Skills LaTeX, Git, Linux